

CTU-Chat: A Hybrid Retrieval-Augmented Generation System for University Student Services

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Abstract. University student-service questions combine exact policy terms, Vietnamese abbreviations, numerical calculations, cohort-specific tuition tables, and academic-program relationships. We present CTU-CHAT, a production-oriented retrieval-augmented generation system for Can Tho University that combines Vietnamese BM25 retrieval, dense vector search, reciprocal-rank fusion, and a multilingual cross-encoder reranker. A parent-child index preserves document context: approximately 400-character child chunks are searched while up to 2,800-character parent sections are supplied to the language model. Metadata filters separate actual tuition, exemption bases, exemption policies, scholarships, student loans, social support, and academic regulations.

An intent-aware LangGraph supervisor rewrites only underspecified follow-ups and routes requests to academic, financial, scholarship, or general agents. Neo4j tools answer program and prerequisite questions; typed tools calculate tuition reductions and scholarship eligibility. PostgreSQL stores parent documents and long-term chat history, Redis stores the recent session buffer, and Qdrant stores the rebuildable vector projection. The current benchmark is a Vietnamese 100-question dataset. Its latest recorded run achieved 91/100 LLM-judged passes (81.35% mean score); financial tool experiments achieved 29/30 end-to-end passes. These measurements are reported as system evidence, not as a claim of superiority over an unreplicated baseline.

Keywords: retrieval-augmented generation · hybrid retrieval · BM25 · reranking · tool use · university chatbot

1 Introduction

University students ask about tuition, exemptions, scholarships, loans, academic programs, prerequisites, and administrative procedures. The answers are distributed across regulations, announcements, fee tables, forms, and program curricula. A useful assistant must therefore retrieve exact numbers and identifiers, preserve table structure, distinguish current from obsolete documents, and avoid replacing missing evidence with a fluent guess.

Pure vector search handles paraphrases but can miss course codes, decision numbers, and Vietnamese policy phrases. Pure lexical search has the opposite

trade-off. A practical student-service system also needs structured lookups and deterministic calculations. We ask the following questions:

- Can BM25 and dense retrieval complement one another on CTU policy and tuition documents?
- Does parent-child chunking improve context preservation for tables and structured Markdown?
- Does cross-encoder reranking with a temporal tie-break change source ranking without hiding the underlying evidence?
- Does intent routing plus typed tool use improve correctness for academic, tuition, scholarship, and loan requests?

The contribution is an auditable end-to-end design rather than a new neural retriever: (1) Vietnamese BM25 with PyVi tokenization and persisted snapshots; (2) Qdrant dense retrieval fused with BM25 by RRF; (3) table-aware parent-child chunking and metadata filtering; (4) a BGE reranker with a bounded date-based tie-break; (5) LangGraph supervision with specialist agents and Neo4j academic tools; and (6) deterministic tuition and scholarship calculations with recorded evaluation results.

2 Related Work

Retrieval-augmented generation (RAG) conditions a generator on external passages and is effective for knowledge-intensive tasks [3]. The CTU setting adds two operational constraints: policy documents are versioned and many answers contain exact values.

BM25 remains a strong lexical baseline for identifiers and terminology [5]. Dense multilingual embeddings handle paraphrases; the implementation uses a local Vietnamese bi-encoder with 768-dimensional vectors. Reciprocal Rank Fusion (RRF) combines ranked lists without requiring calibrated scores [1]. The system consequently keeps BM25 and dense scores separate, fuses ranks, then applies a cross-encoder.

GraphRAG uses explicit relations to retrieve connected evidence [2]. Here the Neo4j projection models programs, blocks, courses, prerequisites, parallel courses, and responsible units. It is used primarily by the academic agent, while policy and financial questions use the hybrid document retriever. This division avoids forcing every question through graph expansion.

ReAct-style agents interleave model decisions and tools [6]. Our agents are bounded: the supervisor chooses ‘academic’, ‘financial’, ‘scholarship’, or ‘general’; financial and scholarship agents receive one calculation tool each; academic agents receive allow-listed Neo4j tools; and the general agent receives no tools. The design is therefore closer to controlled orchestration than open-ended autonomous planning.

Table 1. Implemented capability comparison. A check denotes implementation, not empirical superiority.

System	Retrieval	Control and data support
Dense RAG	Dense vectors	Passage retrieval; no typed business tools
BM25 RAG	Vietnamese BM25	Exact terms; no semantic fusion
CatRAG-style sys-tem [4]	Dense plus cate- gory graph	Category routing and graph enrichment
CTU-CHAT	BM25 + dense + reranker	Intent routing, Qdrant, PostgreSQL, Re- dis, Neo4j, typed tools

3 Problem Formulation

Let a request be $q = (x, i, h)$, where x is the user text, i is the inferred intent, and h is the bounded recent conversation history. The intent space contains tuition, exemption policy, exemption basis, scholarship, student loan, social support, academic program, academic rules, general regulation, calculation, both-fee, and other categories. A routing decision additionally stores an optional academic year and whether it came from the original query or a validated rewrite.

Each document is represented by a parent section p and child chunks $e_1, \dots, e_m$. A retrieval result is admissible only when its metadata matches the selected lane. In particular, inactive records are excluded and actual tuition is not substituted for the exemption basis. For a lane l , the candidate set is

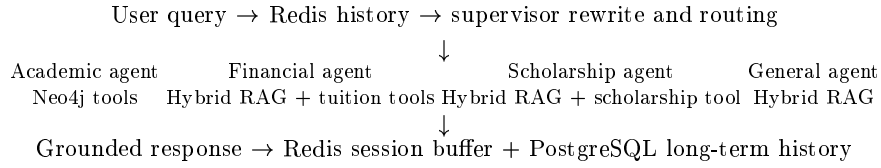
$$\mathcal{A}_l(q) = \{e \mid \text{status}(e) = \text{active}, \text{domain}(e) = \text{domain}_l, \text{content}(e) = \text{content}_l, \text{fee}(e) = \text{fee}_l\}. \quad (1)$$

The system returns an answer y assembled from retrieved parent documents, structured catalog records, or a validated tool result. If the selected lane has no evidence, the answer must say so. Academic-program queries may bypass document retrieval and use Neo4j, while ambiguous tuition queries remain ambiguous until a follow-up or safe rewrite supplies the missing fee type, program, cohort, or training mode.

4 System Architecture

The online request path is a LangGraph state machine:

The supervisor first checks whether the deterministic classifier can settle the request. Only queries classified as ‘other’ or ‘ambiguous_tuition’ are sent to the rewrite model. The candidate rewrite is accepted only if it preserves the business intent and does not introduce unsupported entities, numbers, or conditions. A structured route then selects one of four agents. Academic program questions go directly to the academic ReAct agent; other lanes execute hybrid retrieval before the specialist agent.

**Fig. 1.** CTU-Chat online orchestration path.**Table 2.** Runtime components and implemented responsibility.

Component	Implementation	Responsibility
API	FastAPI	Authenticated ‘/chat’, upload, and history endpoints
Dense index	Qdrant, cosine distance	Child-vector retrieval; 768 dimensions
Lexical index	BM25s + PyVi	Vietnamese tokenization and sparse retrieval
Reranker	‘BAAI/bge-reranker-v2-m3’	Cross-encoder reranking; date tie-break within tolerance
Document store	PostgreSQL	Parent content, metadata, users, and long-term history
Graph	Neo4j 5.20	Programs, courses, prerequisites, and graph tools
Session store	Redis 7	Recent conversation buffer
Generator	Gemini ‘gemini-3.5-flash-lite’	Supervisor and specialist generation

The data path is separated from the authority path. PostgreSQL stores users, chat history, parent documents, and document metadata. Qdrant stores 768-dimensional child vectors. A persisted BM25s snapshot stores child records and parent IDs. Neo4j stores the academic-program graph. Redis keeps the recent session buffer; the API reads the latest five messages and trims the list to the latest fifty entries. All projections can be rebuilt from source documents and structured data.

5 Retrieval, Routing, and Tool Method

5.1 Document ingestion and parent-child chunking

The offline pipeline accepts Markdown produced manually or by the LlamaParse OCR service. Markdown headers are retained in the text and also stored as metadata. Parent sections use a recursive splitter with a 2,800-character limit and 100-character overlap. Each parent is then split into 400-character children with 50-character overlap. A table-aware splitter repeats the table header and the most recent category row in every child, preventing a row from losing its column meaning. Parent content is stored in PostgreSQL; child metadata and embeddings are indexed for retrieval.

5.2 Vietnamese BM25 retrieval

The lexical index uses ‘bm25s’ with the Lucene scoring variant, $k_1 = 1.5$, and $b = 0.75$. PyVi tokenizes Vietnamese text, punctuation is removed, and tokens are lower-cased. Search retrieves up to 15 children (or fewer when the index is smaller), filters metadata, groups matches by parent ID, and keeps the maximum child score for each parent. The index and its record manifest are persisted so that a restart does not silently rebuild a different corpus.

5.3 Dense retrieval and reciprocal-rank fusion

The dense branch uses the configured local ‘bkai-foundation-models/vietnamese-bi-encoder’ when available. Qdrant stores 768-dimensional vectors with cosine distance. For query q , child document e has cosine score

$$s_{\text{dense}}(q, e) = \frac{v_q^\top v_e}{\|v_q\| \|v_e\|}. \quad (2)$$

The dense and BM25 parent lists are fused by

$$s_{\text{RRF}}(p) = \sum_{r \in \{\text{dense}, \text{BM25}\}: p \in r} \frac{1}{60 + \text{rank}_r(p)}. \quad (3)$$

The engine keeps candidates from one branch when the other branch returns no result. It then reranks at most 15 fused parents and returns the top six.

5.4 Cross-encoder reranking and temporal tie-break

The default second-stage model is ‘BAAI/bge-reranker-v2-m3’. It scores each query-parent pair directly. The implementation first sorts by the model score and timestamp, then reorders only documents in the same score bucket, where the score difference is at most 0.05. The newer timestamp wins inside that bucket; a clearly more relevant older document remains ahead. This is a bounded tie-break, not a freshness boost applied to every result.

5.5 Intent routing and agents

The deterministic classifier recognizes tuition, exemption, scholarship, student-loan, social-support, academic-program, academic-rules, and general regulation phrases, including academic-year expressions such as ‘2025-2026’. If a query is underspecified, the rewrite model sees only the previous user question and the current question. A validation step rejects a rewrite that changes intent or adds an entity, number, bank, program, year, or condition.

The LangGraph supervisor emits one of four routes. The academic route invokes the Neo4j ReAct agent. Financial, scholarship, and general routes run the hybrid document retrieval first. Context is labeled with source, academic year, and fee type; if a required lane is empty, the prompt tells the agent not to substitute another lane.

5.6 Structured tools

The financial agent can call ‘`tin_h_toan_hoc_phi`’, which validates a reduction percentage in $[0, 100]$, computes the reduction from the exemption basis, and clips the payable amount at zero. The scholarship agent can call ‘`tin_h_tien_hoc_bong`’, which selects the highest eligible class from GPA and discipline score thresholds and reports the corresponding amount by field. The academic agent uses Neo4j functions for program lookup, prerequisite chains, shared courses, and reverse course-to-program lookup. Tool outputs are plain, inspectable strings; unavailable data produces an explicit error or not-found message instead of a fabricated value.

6 Experimental Setup

6.1 Corpus and evaluation artifacts

The repository contains 50 Markdown policy and tuition documents under ‘`data/markdown`’ and 114 academic-program Markdown files under ‘`data/markdown_graph`’. The active Qdrant collection used by the reranker experiment contained 1,303 points. The primary chat dataset has 100 Vietnamese questions, balanced across four groups: exemption and social support, tuition, student loans, and scholarships (25 questions each). Expected answers and source filenames are stored with the dataset.

The reported chat evaluation used an LLM judge with a pass threshold of 0.55. The latest 100-question run recorded 91 passes, an 81.35% mean score, and no HTTP/API errors. Because this is a repository evaluation rather than a blinded human study, it is reported as an implementation result and not as a generalizable accuracy estimate.

6.2 Tool-calling experiment

The tool dataset contains ten cases each for structured tuition lookup, scholarship calculation, and tuition-reduction calculation. The experiment checks path/tool selection, argument validity, result correctness, and end-to-end case success. Selection passed 30/30, arguments 29/30, results 30/30, and end-to-end cases 29/30.

6.3 Reranker experiment

The A/B test used the same 100-question dataset. The evaluated collection alias was

`ctu_scholarship_docs_current`.

Eighty questions used RAG retrieval and twenty used an exact tuition-catalog bypass. On the 80 active questions, the expected source appeared in the returned documents for all three variants: plain cross-encoder, tie-break tolerance 0.05, and tie-break tolerance 0.02. Mean expected-source rank was 1.18, 1.27, and 1.24, respectively. Thus the test demonstrates stable source coverage, not a measured quality gain from the tie-break.

6.4 Additional retrieval and answer-quality metrics

The next evaluation will isolate the retrieval paths already present in ‘AdvancedChunkingEngine’: ‘dense_only’, ‘sparse_only’ (BM25), ‘hybrid_rrf’, and ‘hybrid_rrf_rerank’ with the production cross-encoder. The current online path is the last variant; the first three are controlled ablation modes that must disable the other branch or reranker in the evaluation runner. For each document-retrieval configuration we will report precision at five, Recall at five, and mean reciprocal rank:

$$\begin{aligned} P@5 &= \frac{|Rel \cap Top_5|}{5}, & Recall@5 &= \frac{|Rel \cap Top_5|}{|Rel|}, \\ MRR &= \frac{1}{N} \sum_{q=1}^N \frac{1}{rank_q^{first}}. \end{aligned} \tag{4}$$

Here Rel is the set of human-labelled relevant sources and $rank_q^{first}$ is the rank of the first relevant result for query q .

End-to-end answer quality will be evaluated with RAGAS-style LLM-as-a-judge scores: Answer Relevance, Context Recall, Context Precision, and Answer Correctness. The repository already provides ‘evaluate_ragas.py’ and ‘score_ragas.py’ for these measures. We will use the same question, answer, context, and ground-truth records for all retrieval configurations, report the mean and per-domain support count, and retain failed or missing scores instead of converting them to successful cases. The four metric values shown in the planned result table remain pending until this run is completed.

6.5 Graph and agent ablations

Graph and orchestration are evaluated separately from document retrieval. The academic route uses Neo4j tools for program lookup, prerequisite chains, shared courses, and reverse course lookup; it does not return Qdrant or BM25 passages. The planned graph evaluation therefore reports program lookup accuracy, prerequisite-chain exactness, shared-course set F1, and tool-call validity. The routing evaluation compares a fixed route with the supervisor route and reports intent accuracy, selected-agent accuracy, unauthorized-tool rate, and end-to-end answer metrics. The final full-system row combines hybrid retrieval, reranking, supervisor routing, specialist agents, memory, and typed tools.

All graph, routing, and full-system result cells remain ‘pending’ until the corresponding controlled runs are executed. This prevents a document-retrieval metric from being incorrectly interpreted as evidence for graph or agent quality.

The existing implementation metrics are pass rate, mean judge score, source hit, expected-source rank, selection/argument/result accuracy, and end-to-end tool success. Reproduction requires the dataset snapshot, Qdrant collection, BM25 manifest, model identifiers, reranker tolerance, environment variables, and exact log file. The current generator is Gemini ‘gemini-3.5-flash-lite’; the rewrite model is ‘gemini-3.1-flash-lite’ with temperature 0.0. Docker services are Qdrant 1.18.1, Redis 7, PostgreSQL 15, and Neo4j 5.20.

Table 3. Recorded implementation evaluations.

Evaluation	Numerator/denominator	Result	Interpretation
Chat judge pass rate	91/100	91.00%	Latest 100-question run
Chat mean judge score	–	81.35%	Judge score, not exact accuracy
Chat API errors	0/100	0	HTTP/API layer
Tool path selection	30/30	100.00%	Three tool families
Tool argument validity	29/30	96.67%	One argument mismatch
Tool result correctness	30/30	100.00%	Deterministic checks
Tool end-to-end cases	29/30	96.67%	Isolated sessions
Reranker source hit	80/80	100.00%	Each of three variants

7 Results

7.1 Chat quality by domain

The latest logged run passed 25/25 exemption and social-support questions (100%), 23/25 tuition questions (92%), 20/25 student-loan questions (80%), and 23/25 scholarship questions (92%). The aggregate is therefore dominated by policy lookup success and does not establish equal performance across domains.

7.2 Reranking and source coverage

The reranker A/B experiment found the expected source in the returned documents for all 80 RAG-active questions under plain scoring and both temporal tie-break tolerances. Plain scoring had the best mean expected-source rank (1.18), followed by tolerance 0.02 (1.24) and 0.05 (1.27). The tie-break changed some ordering but did not improve source-hit coverage in this test.

7.3 Tool behavior

Structured tuition lookup, scholarship calculation, and tuition-reduction calculation all achieved 10/10 result checks. The only end-to-end failure was associated with argument selection in the scholarship group. The evidence supports using typed tools for arithmetic, while routing and argument generation remain the main error surface.

7.4 Planned retrieval and RAGAS evaluation

Table 4 defines the additional experiment requested for the next run. Values are intentionally left as pending measurements; no comparison is inferred before all configurations use the same labelled queries.

The corresponding answer-quality table is:

These four columns measure complementary failure modes: whether the answer addresses the question, whether the retrieved context covers the reference facts, whether relevant context is ranked early, and whether the final answer

Table 4. Planned retrieval metrics for the next controlled run.

Configuration	P@5	Recall@5	MRR
'dense_only'	pending	pending	pending
'sparse_only'	pending	pending	pending
'hybrid_rrf'	pending	pending	pending
'hybrid_rrf_rerank'	pending	pending	pending

Table 5. Planned RAGAS answer-quality metrics; all values are pending.

Configuration	Answer Relevance	Context Recall	Context Precision	Answer Correctness
Dense-only	pending	pending	pending	pending
Sparse-only	pending	pending	pending	pending
(BM25)				
Hybrid RRF	pending	pending	pending	pending
Hybrid RRF + reranker	pending	pending	pending	pending

matches the reference answer. RAGAS scores will be reported with the judge model, prompt configuration, failed-case count, and per-domain sample size.

For 'graph_only', the graph/path column contains program lookup accuracy, prerequisite-chain exactness, and shared-course F1. For the routing rows, the tool column includes unauthorized-tool rate and argument validity. The answer metrics column contains Answer Relevance, Context Recall, Context Precision, and Answer Correctness where a generated answer exists.

7.5 What the results do not show

The repository does not yet contain the controlled P@5/Recall@5/MRR comparison, the graph/tool ablation, the routing-agent comparison, or the four-metric RAGAS table described above. It also does not contain a statistically powered ablation of parent-child chunking or the Neo4j academic tools. Accordingly, no claim that hybrid retrieval, reranking, or multi-agent routing is superior is made here.

8 Discussion, Limitations, and Responsible Use

The implementation results show a useful but uneven system. Exemption and social-support questions were strongest in the latest run, while student-loan questions were weakest. This pattern is consistent with a corpus containing highly structured policy phrases and several overlapping loan regulations, but the log does not prove the cause. The reranker experiment shows that a temporal tie-break can preserve source coverage, yet its mean rank was worse than plain cross-encoder scoring on this sample.

Table 6. Planned graph and agent ablations; all values are pending.

Variant	Route accuracy	Tool validity	Graph/path metric	Answer metrics
‘graph_only’	n/a	pending	pending	pending
‘fixed_route’ + hybrid	pending	pending	n/a	pending
‘supervisor’ + hybrid	pending	pending	n/a	pending
‘full_system’	pending	pending	pending	pending

The design separates retrieval from deterministic business logic. The tuition catalog bypass is appropriate for exact program/cohort lookups; the hybrid retriever supplies explanatory policy context; and tools perform arithmetic after the model selects validated inputs. Neo4j is valuable for prerequisite and shared-course relations, but it should not be described as replacing the document RAG path for financial or policy questions.

Important limitations remain. The main chat benchmark is Vietnamese only and uses an LLM judge. Source-hit ranking is not answer faithfulness, and no paired human adjudication or confidence interval is available. The corpus contains multiple versions and inactive records, so metadata quality directly affects retrieval. Live OCR, Gemini, Qdrant, Redis, PostgreSQL, and Neo4j services add operational failure modes. The rewrite guard is heuristic and the current session memory is bounded by list length rather than a learned memory policy.

The assistant is decision support, not the final authority for academic or financial decisions. Responses should expose the source document and relevant year where available. Student records, credentials, and personal identifiers must not enter evaluation traces without consent and retention controls.

9 Conclusion

We documented CTU-Chat, a university student-service assistant that combines Vietnamese BM25, dense Qdrant retrieval, RRF, parent-child context recovery, cross-encoder reranking, intent routing, Neo4j academic tools, and typed financial calculations. The latest repository evaluations recorded 91/100 LLM-judged chat passes and 29/30 end-to-end tool cases. Reranker A/B tests maintained 100% expected-source coverage on 80 RAG-active questions, but did not demonstrate a tie-break gain.

The central engineering lesson is separation of concerns: exact catalog lookups and arithmetic belong to structured components, semantic and lexical retrieval complement one another, and graph queries should be used where the question is relational. A frozen bilingual benchmark, human evaluation, paired retrieval ablations, and versioned latency measurements are required before making stronger comparative claims.

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